

Some questions on circular motion - centripetal force including vertical circles

1. a.

A cyclist leans over while cornering to shift their center of mass towards the center point of the corner. As the distance to the center of the corner decreases, the force needed to successfully complete the corner at some speed decreases proportional to the distance to the center.

b. The centripetal force F_c can be found by

$$\begin{aligned} F_c &= \frac{mv^2}{r} \\ &= \frac{80 \cdot 21^2}{32} \\ &= 1100 \text{ N} \end{aligned}$$

c. Towards the center point of the corner.

d. $R \cos \theta = 785 \text{ N}$, and $R \sin \theta = 1100 \text{ N}$

$$\therefore \tan \theta = \frac{1100}{785}; \text{ so } \theta = 55^\circ.$$

$$\text{So } R = \frac{785}{\cos(55)} = 1350 \text{ N}.$$

e. Friction

2.

a.

$$\begin{aligned} F_c &= \frac{mv^2}{r} \\ 0.9mg &= \frac{mv^2}{r} \\ r &= \frac{31^2}{0.9 \cdot 9.81} \\ &= 109 \text{ N} \end{aligned}$$

b. The car may roll or the back wheels may skid out, causing the driver to lose control.

c. The F1 cars provide downforce, which increases the vertical force on the tires. This causes the maximum frictional force that the tires can exert on the ground to be much higher, so the cars can move faster around tighter corners.

3. a. $\omega = \frac{2\pi}{30 \cdot 60} = 3.49 \cdot 10^{-3} \text{ rad s}^{-1}$

b. Assuming that 135m is the height of the wheel,

$$\begin{aligned} F_c &= \left(65 \cdot (3.49 \cdot 10^{-3})^2 \cdot 67.5 \right) \\ &= 0.05 \text{ N} \end{aligned}$$

c. At the highest point: gravity.

At the lowest point: the reaction force of the car on the person.

Halfway: the frictional force of the person's shoes against the floor of the car.

d. At the top, since at the bottom the reaction force = $mg + F_c$, where F_c is the centrifugal force and mg is the weight.

4. a. We know that $F_c \geq mg$.

$$\frac{mv^2}{r} \geq mg$$

$$\frac{v^2}{r} \geq g$$

$$v \geq \sqrt{gr}$$

$$\geq \sqrt{g \cdot 3.4}$$

$$\geq 5.78\text{ms}^{-1}$$

So $\min(v) = 5.78\text{ms}^{-1}$

b.